

SmartCare v0.3 - Domain Model Workbook

Week 6 student resource

Requirement-to-Concept Trace

Requirement	Concept	State/behaviour	Decision
FR-01 Create patient record with unique ID	Patient	State: patient_id, name	Class
FR-02 Search patient by ID or name	Patient	Behaviour: matches(query)	Behaviour on Patient
FR-03 Store practitioner records	Practitioner	State: practitioner_id, name	Class
FR-04 Create appointment (patient, practitioner, date, time)	Appointment	State: patient, practitioner, date, time	Class
FR-05 Reject a second booking for the same practitioner, date and time	Appointment / AppointmentBook	Behaviour: conflicts_with(other); add() rejects conflicts	Rule needs to see all appointments, so it sits in an optional AppointmentBook
FR-06 Reject appointment for unknown patient or practitioner	Appointment	Validation when created	Behaviour, not a class
FR-07 Every appointment has a status	Appointment	State: status (Booked, Cancelled, Completed)	Attribute, not a class
FR-08 Staff can cancel an appointment	Appointment	Behaviour: cancel()	Behaviour, not a class
FR-09 Cancelled appointments stay in the history	Appointment	Record kept with status Cancelled	Provisional rule, to confirm with client
FR-10 Show a practitioner's appointments for a date	Practitioner	Behaviour: appointments_on(date)	Behaviour on Practitioner

CRC Cards

Patient

Responsibilities	Collaborators
Hold patient ID and name	(none)
Be found by ID or name and provide its details	Appointment

Practitioner

Responsibilities	Collaborators
Hold practitioner ID and name	(none)
Provide its appointments and availability for a date	Appointment

Appointment

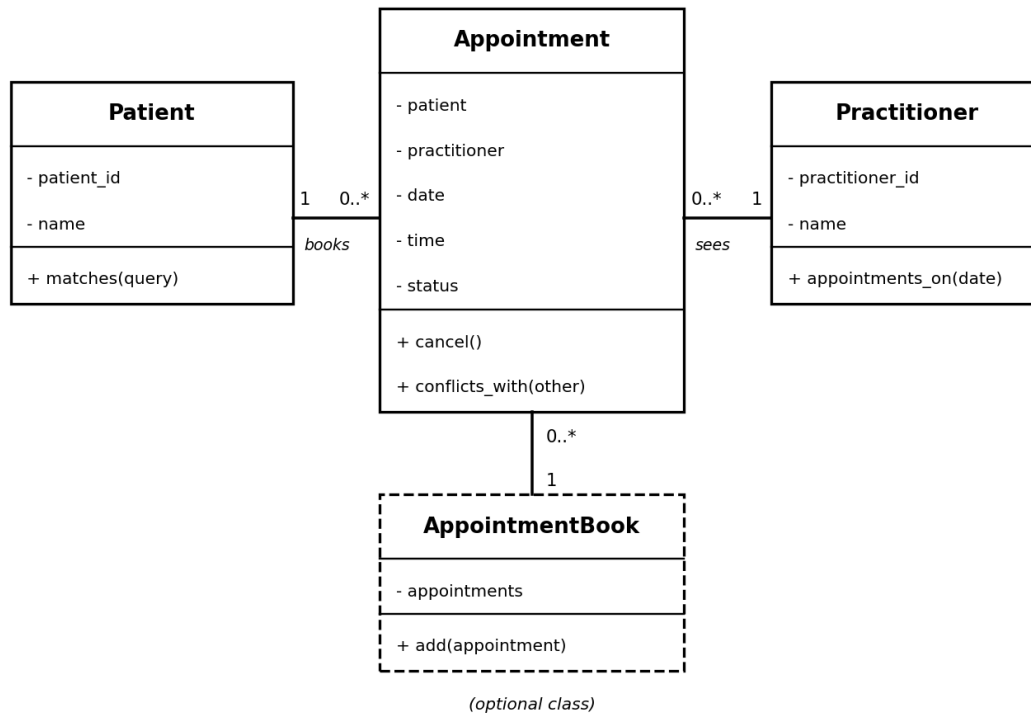
Responsibilities	Collaborators
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Hold its patient, practitioner, date, time and status	Patient, Practitioner
Be cancelled while staying in the history; detect a clash with another appointment	AppointmentBook

Optional class

Responsibilities	Collaborators
AppointmentBook (optional): hold all appointments and reject practitioner clashes	Appointment
Provide appointments by practitioner and date	Appointment, Practitioner

UML Class Diagram



Design Rationale

Patient, Practitioner and Appointment are classes because each is a real clinic concept with its own data and behaviour, and each traces to FR-01 to FR-10. Name, Status and Cancellation are not classes: name and status are attributes, and cancellation is an operation on Appointment (FR-08). Patient and Practitioner each have a one-to-many association with Appointment, because an appointment involves exactly one of each and both exist independently. The no-double-booking rule (FR-05) cannot be decided by one appointment alone, so the optional AppointmentBook holds the list and checks for clashes. AppointmentBook is optional and would be dropped if the rule is enforced elsewhere. The manager, controller, notification and scheduling classes suggested by AI have no supporting requirement.

AI Design Review Record

AI suggestion	Evidence	Decision	Reason	Model change
Add an AppointmentBook class to check practitioner clashes	FR-05	Accepted	FR-05 needs something that can see all appointments	Added AppointmentBook as an optional class
Make Cancellation a separate class	FR-08, FR-09	Rejected	FR-08 only needs a cancel() operation and a status	None
Make Status a separate class	FR-07	Modified	FR-07 needs three values, not a class	Kept Status as an attribute limited to Booked, Cancelled or Completed
Add NotificationManager	None (reminders were out of scope)	Rejected	No requirement, and reminders were out of scope	None
Add ClinicController that owns all objects	None	Rejected	No evidence, and it would become an overloaded class	None

Reflection

The hardest decision was where the no-double-booking rule should live. One appointment can't see the others, so I added an optional AppointmentBook to check clashes.

The AI over-designed by suggesting manager, controller, notification and Cancellation classes with no requirements behind them, so I rejected them.

My final choices were backed by FR-01 to FR-08. Anything unconfirmed, like keeping cancelled appointments, stayed marked provisional.